AWS VPC Beginner — Common Mistakes & Solutions

Task	Beginner Mistake	Solution
VPC	Wrong CIDR, e.g. `10.0.0.0/24`	Use the required `10.0.0.0/16`
Subnet	Public and private subnet have overlapping CIDRs	Use different ranges, e.g. `10.0.1.0/24` and `10.0.2.0/24`
Availability Zone	Public and private subnet are in different AZs when the test requires same AZ	Select the **same AZ** for both
Public Subnet	Auto-assign Public IP disabled	Enable **Auto-assign IPv4 address**
Internet Gateway	IGW created but not attached	**Actions → Attach to VPC**
Public Route Table	No Internet route	Add `0.0.0.0/0 → Internet Gateway`
Public Route Table	Route table not associated with Public Subnet	Go to **Subnet associations → Edit → select Public Subnet**
Private Route Table	Private subnet associated with Public Route Table	Create a separate **Private Route Table**
NAT Gateway	NAT Gateway created in Private Subnet	NAT Gateway must be in the **Public Subnet**
NAT Gateway	No Elastic IP	Allocate an **Elastic IP** and attach it to the NAT Gateway
Private Route	Private subnet has no Internet route	Add `0.0.0.0/0 → NAT Gateway`
EC2	EC2 launched in Private Subnet when testing SSH from laptop	Launch the test EC2 in the **Public Subnet**
EC2 Public IP	EC2 has no public IP	Enable **Auto-assign Public IP** or assign an Elastic IP
Security Group	SSH port 22 not allowed	Allow **SSH (22)** from **My IP**
Security Group	HTTP port 80 not allowed	Add **HTTP (80)** from `0.0.0.0/0`

SSH	Wrong username	Ubuntu → `ubuntu`, Amazon Linux → `ec2-user`
SSH	`.pem` permission error	Run `chmod 400 key.pem`
Apache	Apache not installed	`sudo apt install apache2 -y`
Apache	Apache not running	`sudo systemctl start apache2`
Website	Using HTTPS instead of HTTP	Try `http://<Public-IP>`
Website	Apache works locally but not from browser	Check **Security Group port 80** and public route
NAT	NAT Gateway shows Pending	Wait until it becomes **Available**
Private EC2	Trying to SSH directly from Internet	Private EC2 needs a **bastion/public host or SSM**
Route table	Route points to wrong gateway	Check that Public → **IGW**, Private → **NAT Gateway**

Issues

1) We can't connect directly to the private server because it doesn't have a private ip address
2) Ping https://www.google.com not working
3) Task14 Im using mac so it doesnt support SCP to transfer key to the public server

Troubleshooting

1) We can connect to the private server through creating public server
2) Or we can create VPC endpoint to connect the private server directly
3) Although i cant connect so ive curl the private sever by creating vpc endpoint
4) Instead of ping i've used curl command that gives a html code of google.com with the wordings with GOOGLE INC